ECHO-7

Integrated vision-guided bay search and docking

DATASHEET / 04 NOV 2026

Design E7070004 / CCTL E7070001

SYSTEM / CURRENT IMPLEMENTATION

DE1-SoC + OV7670 + UGV02 | Cyclone V 5CSEMA5F31C6 | Production top: **echo7IntegratedTop**. One SOF selects real vision, synthetic-vision chassis debugging or integrated operation. Physical IR and J3 sonar always remain in the motion safety chain.

MODE	INPUT / OUTPUT
INTEGRATED (power-up default)	Real camera; explicitly started tasks only. No host heartbeat required.
VISION_ONLY	Real tracking/calibration; wheel UART remains zero, even without a valid sonar reading.
CHASSIS_INJECT	JTAG observations at 25 Hz through the real C controller; physical IR/J3 remain active.

CONTROLS & VISUAL WORKFLOW

SW0-SW5: one-hot bay 1-6; approximately 20 ms stability. No selection/conflict cannot move. Selection changes cancel the task; valid selection tracks before ARM.

SW6/SW7: candidate / centre RGB-HSV diagnostics. All 16 candidates appear on two 8-slot pages, about 2 s/page. **SW8:** perspective bypass. **SW9:** raw display, cancel motion.

Red1: colour calibration. **Red2:** reselect the current switches and request a new task. Uncalibrated mode waits for a new 3-frame target lock. Remove the card for 3 healthy frames before presenting it for 3 frames; a resident card does not start at reset.

IR A (0x0F): latched stop. **POWER (0x12):** first fresh press releases an IR stop; a second press starts. Repeats/held POWER do not start twice. **KEY0:** full reset.

MOTION & STOP CONDITIONS

Search: opposite logical wheel requests at 4 cm/s. First valid observation stops rotation; await a complete zero-speed JSON, then 200 ms conservative dwell and 3 new independent frames.

Alignment: U=110-130 centred (U range 0-239). **Approach:** 5 cm/s above 500 mm; 3 cm/s at 350-500 mm. **Dock:** 350 mm with an aligned, locked target.

Guard: distance <=250 mm, invalid or stale data stops motion. HOLD recovery needs 3 new safe samples >=300 mm. DOCKED/NOT_FOUND stay disarmed until a new start.

Freshness: visual reports about 150 ms; injection lease 500 ms (host renewal about 100 ms). Lost lease cancels ARM; renewed heartbeat does not restart. UART: 115200 8N1, complete JSON, zero-speed priority.

SCAN_UNCALIBRATED: no verified heading feedback or three-trial floor calibration. The default 10 s is a bounded injection-debug budget, not a measured 360 degrees. INTEGRATED blocks unseen-target scanning until calibrated; an already locked visible target may still dock. Search has a 120 s total task limit.

EVIDENCE / CURRENT BUILD & HARDWARE

CHECK	RESULT / LIMIT
Directed control & protocols	PASS: control 254 checks; CCTL 58 checks. Complete UART bytes decoded.
Earlier pin-level simulation	PASS on pre-red2 RTL: 9 sonar model measurements; 3 moving / 6 zero UART lines. NEC repeat/release checked; 97.734 s runtime.
Pixel-to-UART system	PASS: 12 fixed-rate 25 fps frames through perceptionCore, real C and UART. Synthetic pixels; not a live-camera acceptance.
Course R6	PASS 219.556 s for E7070004; Verilator 5.050 clean copy, jobs=4; 12 PASS + deliberate mutant CAUGHT. Meets the 285 s user limit.
Quartus fit / timing	PASS E7070004: compile 1,121.141 s + gate 25.868 s. ALM 29,576/32,070 (92%); M10K 357/397; DSP 51/87; 41,000 registers. Worst setup +1.633 ns / hold +0.118 ns; unconstrained 0; 0 errors / 92 warnings.
Current-board red2 / vision	E7070004 red2 tag 3->4, new lock and ARM/ALIGN (+4/-4 cm/s, J3 468 mm). User confirmed physical start and stop when blue1 left view. 13 no-heartbeat samples stayed armed, with transient LOST/ACQUIRE. Final: VISION_ONLY, disarmed, zero; user confirmed stopped. STOP's separate mechanical timing was not measured. Blue1 q255, processing 47.652 ms; flags=3 retained.
Earlier wheel / IR checks	Earlier E7070002: synthetic left/centre/right, lease loss/no automatic restart, bounded NOT_FOUND, and real IR A/POWER1/POWER2 passed. User confirmed wheel actions and both sides forward for logical +3/+3 cm/s. These are prior-build checks.

Unmeasured: floor 360-degree repeatability, six-bay closed-loop trials, physical docking error, scene-to-UART latency and SW8 comparison. Transient visual loss/reacquisition and historical frame30321 blue6 fragmentation remain limitations.

Current SOF: firmware/E7070004/PartBDebug.sof; 6,706,154 bytes. Original build: 20261104T111202_124373Z_build. Programmed-board ID E7070004 verified.

Reproduction: README.md / README_EN.md; docs/REPAIR_RESULT.md; tools/render_integrated_datasheet.py. No direct wheel or synthetic-distance override.